

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : MCQ Test (Low)
Assessment Tool Weightage: 30%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	5
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	5
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	5
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	5
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	5
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	5
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	5

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	5
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	5
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	5

Code of Conduct:

I Shaik Ayesha Tabassum(192525074) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S.Ayesha Tabassum

RUBRICS FOR CALCULATION:

MCQ					
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding ; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1.Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies

(c) All attributes must be atomic

Explanation: In **First Normal Form (1NF)**, every attribute must contain **atomic (single, indivisible) values**. Multi-valued or repeating groups are not allowed.

2.A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies

(b) Has no partial dependencies on primary key

Explanation: A relation is in **2NF (Second Normal Form)** if:

1. It is already in **1NF**, and
2. It has **no partial dependency** every non-prime attribute must depend on the **whole primary key**, not just part of a composite key.

3.Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF

(d) 4NF

Justification:

Fourth Normal Form (4NF) specifically deals with **multi-valued dependencies (MVDs)**.

A relation is in **4NF** if:

- It is already in **BCNF**, and
- It has **no non-trivial multi-valued dependencies** unless the determinant is a **super key**.

4.BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF

(c) 3NF

Justification:

BCNF (Boyce-Codd Normal Form) is a **stricter version of 3NF**.

- **1NF** → Ensures atomic values.
- **2NF** → Eliminates partial dependencies.
- **3NF** → Eliminates transitive dependencies.
- **BCNF** → Strengthens 3NF by requiring **every determinant to be a super key**.

5. Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other

(b) X determines Y

Justification:

A **functional dependency (FD)** $X \rightarrow Y$ means that the value of **X uniquely determines the value of Y**.

In other words, if two rows have the same value of **X**, they must have the same value of **Y**.

6. Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly

(d) Retrieval anomaly

Justification:

The main anomalies are **Insertion, Deletion, and Update**. **Retrieval anomaly** is not a standard database anomaly.

7. A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference

(b) Any set of attributes that uniquely identifies tuples

Justification:

A **superkey** is any set of one or more attributes that can **uniquely identify each row (tuple)** in a relation.

8. If $R(A,B,C)$ has FD: $A \rightarrow B$ and $B \rightarrow C$, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined

(c) Transitively dependent on A

Justification:

Given:

- $A \rightarrow B$
- $B \rightarrow C$

So, **A determines B**, and **B determines C**. Therefore, **C depends on A through B**, which is called a **transitive dependency**.

9. 5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form
(c) Project-Join Normal Form

Justification:

5NF (Fifth Normal Form) is also called **Project-Join Normal Form**. It deals with **join dependencies** and ensures that a relation cannot be further decomposed without losing information

10. Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist

(a) No data is lost during decomposition

Justification:

A **lossless-join decomposition** ensures that when decomposed tables are **joined back**, we get the **original relation without losing or adding incorrect data**.